E-COMMERCE CUSTOMER CHURN & RETENTION ANALYTICS

Comprehensive Analytics, Machine Learning & Multi-Platform Visualization
(Python, Power BI, and Tableau Public)

Submitted by:

Dona Abraham

Roll Number: 2501722037003

MCA / Data Analysis & Visualization (DAV)

TABLE OF CONTENTS

1. Introduction 3
2. Problem Statement 4
3. Objectives 4
4. Dataset Description 5
    4.1 Dataset Size 5
    4.2 Attributes 6
    4.3 Target Variable 6
    4.4 Dataset Characteristics 7
5. Data Preprocessing 7
6. Exploratory Data Analysis & Analytical Techniques 10
7. Machine Learning (Logistic Regression) 13
8. Python Visualizations 15
    8.1 Visual Interpretation and Retention Insights 21
9. Power BI Dashboard 22
10. Tableau Dashboard 23
11. Excel AI Analysis 25
12. Comparison of Tools 26
13. Findings 27
14. Conclusion 28
15. References 28
Appendix A: Complete Python Analytics Code (Spyder 6) 29

1. Introduction

Customer retention is an important factor in the success of an e-commerce business. In a competitive online market, businesses need to not only attract new customers but also maintain good relationships with existing customers. Losing existing customers can result in reduced revenue and increased customer acquisition costs. Customer churn refers to the situation where a customer stops purchasing products or interacting with an e-commerce platform. Therefore, churn analysis is useful because it converts customer activity into measurable evidence that can be used to understand where retention problems are occurring.

This project focuses on analyzing customer behavior and predicting whether a customer is likely to churn. A robust enterprise dataset of 15,000 customer records is evaluated across key customer-related attributes including age, gender, tenure, preferred payment mode, order count, cashback amount, and satisfaction score to uncover transaction patterns associated with customer churn.

Data Analytics techniques are used to understand customer characteristics, purchasing behavior, and retention patterns. Data preprocessing techniques such as removing duplicate records, handling missing values through statistical imputation, encoding categorical variables, scaling numerical data, and removing unnecessary ID columns are applied to improve the quality and consistency of the dataset.

Different visualizations are developed to understand the relationship between customer characteristics and churn. The analysis examines churn vs. retained customer ratios, churn by payment method, order count distributions, and tenure vs. churn relationships. These visualizations help identify actionable patterns in customer behavior.

The project uses Logistic Regression, a machine learning classification algorithm, to predict customer churn. The model classifies customers into two categories: Retained (0) and Churned (1). Model performance is evaluated using measures such as accuracy, precision, recall, F1-score, and a confusion matrix. The insights obtained from the analysis assist e-commerce businesses in formulating targeted retention strategies such as personalized loyalty rewards, onboarding discounts, friction-free payment methods, and responsive customer service.

Overall, this project demonstrates how data analytics and machine learning can be applied to customer data to understand churn behavior, predict customer attrition, and support data-driven decision-making in e-commerce businesses.

2. Problem Statement

Project Scope: Title: E-Commerce Customer Churn & Retention Analytics

The project aims to address the following problems:

- Identify customer churn by analyzing customer behavior and purchase patterns.
- Determine factors influencing churn, such as tenure, order count, payment mode, cashback amount, and satisfaction score.
- Differentiate retained and churned customers based on the available customer data.
- Identify customers at risk of leaving the e-commerce platform.
- Analyze customer retention patterns to understand why some customers continue using the platform while others leave.
- Support targeted retention strategies by providing meaningful insights about customer behavior.
- Reduce potential revenue loss by helping businesses take appropriate action to retain existing customers.

3. Objectives

- To analyze customer data across 15,000 records to understand purchasing behavior and demographic trends.
- To identify the key drivers of customer churn, including payment mode, tenure, and satisfaction ratings.
- To clean and preprocess the customer dataset by handling duplicate entries, imputing missing values, and scaling features.
- To perform exploratory data analysis and answer five essential analytical questions on customer retention.
- To build and validate a Logistic Regression machine learning model to predict customer churn.
- To classify customers accurately into Retained (0) and Churned (1) categories.
- To visualize patterns and relationships using professional Python-based charts and statistical plots.
- To formulate targeted retention strategies that assist e-commerce businesses in reducing customer attrition and revenue loss.

4. Dataset Description

The dataset used for this case study contains records related to customer account history, transactional activities, engagement metrics, and feedback scores on an e-commerce platform. It provides information about customer demographics, loyalty duration, order volume, promotional rewards, and churn status.

The dataset contains 15,000 customer records and 11 attributes. These attributes provide sufficient detail to study customer retention patterns and build a predictive classification model.

4.1 Dataset Size

Attribute	Description
Dataset Name	E-Commerce Customer Churn Dataset
Number of Records	15,000
Number of Attributes	11
Dataset Type	Structured Tabular Dataset
File Format	CSV (Comma-Separated Values)
Domain	E-Commerce / Customer Intelligence
Target Variable	Churn (0 = Retained, 1 = Churned)
Machine Learning Task	Binary Classification

4.2 Attributes

Attribute	Description
CustomerID	Unique identification number of the customer (e.g., CUST-10000)
Age	Chronological age of the customer in years (Range: 18 - 72)
Gender	Gender of the customer (Male, Female)
CityTier	City dwelling classification (Tier 1, Tier 2, Tier 3)
Membership	Loyalty membership tier (Bronze, Silver, Gold, Platinum)
Tenure	Duration for which customer has been with platform in months (0 - 60)
PreferredPaymentMode	Primary payment method (Credit Card, Debit Card, UPI, E-Wallet, COD)
OrderCount	Total cumulative orders placed by the customer (Range: 1 - 25)
CashbackAmount	Average promotional cashback received by the customer in USD ($)
SatisfactionScore	Customer self-reported rating on delivery and service (Scale: 1 to 5)
Churn	Indicates whether the customer has churned (0 = Retained, 1 = Churned)

4.3 Target Variable

The target variable selected for this project is Churn. It represents whether a customer continues to transact on the platform or has abandoned it:

- 0 - Retained Customer: The customer continued using the platform / purchasing products. They have not left the business.
- 1 - Churned Customer: The customer stopped purchasing products and discontinued platform activity.

Because the target variable has two distinct binary outcomes, the project treats this as a Binary Classification problem using Logistic Regression.

4.4 Dataset Characteristics

The dataset contains a balanced mixture of numerical, categorical, and identifier attributes:

- Numerical Variables: Age, Tenure, OrderCount, CashbackAmount, SatisfactionScore.
- Categorical Variables: Gender, CityTier, Membership, PreferredPaymentMode.
- Identifier Variable: CustomerID

5. Data Preprocessing

Data preprocessing was performed using Python and the Pandas library to prepare the raw dataset for analysis and machine learning. The dataset was examined for duplicates, missing values, data types, and scaling requirements. Each preprocessing step has a specific purpose: missing-value treatment prevents incomplete records from distorting calculations, duplicate removal prevents repeated customers from being counted twice, categorical encoding converts text categories into model-readable values, and removal of CustomerID prevents an identifier from acting as a predictive feature.

Loading the Dataset

The dataset was loaded into Python using Pandas:

```
import pandas as pd
df = pd.read_csv("ecommerce_customer_churn.csv")
print("Dataset Shape:", df.shape)
print(df.head(5))
```

Output: First 5 Sample Records:

CustomerID	Age	Gender	Payment Mode	Tenure	Orders	Cashback	CSAT	Churn
CUST-10000	42	Male	Debit Card	11.0	8	$221.85	5.0	0 (Retained)
CUST-10001	40	Female	E-Wallet	4.0	4	$182.86	4.0	1 (Churned)
CUST-10002	18	Female	COD	0.0	6	$229.94	4.0	0 (Retained)
CUST-10003	33	Female	E-Wallet	11.0	3	$177.60	5.0	0 (Retained)
CUST-10004	48	Male	Debit Card	3.0	8	$196.89	5.0	0 (Retained)

Checking Missing Values

Missing values were checked across all attributes before performing analysis:

```
print("Missing values in dataset:")
print(df.isnull().sum())
```

Attribute	Missing Values Count	Imputation Strategy
CustomerID	0	None Required (Identifier)

Age	0	None Required
Gender	0	None Required
CityTier	0	None Required
Membership	0	None Required
Tenure	355	Median Imputation (Robust to skewness)
PreferredPaymentMode	0	None Required
OrderCount	225	Median Imputation (Robust to skewness)
CashbackAmount	286	Mean Imputation ($194.99)
SatisfactionScore	205	Mode Imputation (4.0 Stars)
Churn	0	None Required (Target Variable)

Missing values were imputed using domain-appropriate statistical methods: Tenure and OrderCount were imputed with their Median values (robust against right-skew); CashbackAmount was imputed with the Mean ($194.99); SatisfactionScore was imputed with the Mode (4.0).

Checking Duplicate Records

```
print("Duplicate records detected:", df.duplicated().sum())
df = df.drop_duplicates()
print("Dataset Shape After Removing Duplicates:", df.shape)
```

Output: Duplicate records detected: 300
Dataset Shape After Removing Duplicates: (15000, 11)

Checking Data Types

```
print(df.dtypes)
```

Verified Types: CustomerID: object | Age: int64 | Gender: object | CityTier: object | Membership: object | Tenure: float64 | PreferredPaymentMode: object | OrderCount: float64 | CashbackAmount: float64 | SatisfactionScore: float64 | Churn: int64

Encoding Categorical Variables

Categorical variables (Gender, CityTier, Membership, PreferredPaymentMode) were converted into numerical dummy columns using One-Hot Encoding:

```
categorical_cols = ['Gender', 'CityTier', 'Membership', 'PreferredPaymentMode']
df_encoded = pd.get_dummies(df.drop(columns=['CustomerID']), columns=categorical_cols,
drop_first=True)
print("Encoded Dataset Shape:", df_encoded.shape)
print("Columns After Encoding:\n", list(df_encoded.columns))
```

Output: Encoded Dataset Shape: (15000, 16)
Columns: ['Age', 'Tenure', 'OrderCount', 'CashbackAmount', 'SatisfactionScore', 'Churn', 'Gender_Male',

'CityTier_Tier 2', 'CityTier_Tier 3', 'Membership_Gold', 'Membership_Platinum', 'Membership_Silver', 'PreferredPaymentMode_Credit Card', 'PreferredPaymentMode_Debit Card', 'PreferredPaymentMode_E-Wallet', 'PreferredPaymentMode_UPI']

6. Exploratory Data Analysis & Analytical Techniques

Exploratory Data Analysis was performed using 5 core statistical techniques: Mean, Median, Mode, Standard Deviation, and Correlation Analysis. These measures answer different questions: the mean summarizes the typical numerical value, the median shows the middle observation and is useful when distributions are skewed, the mode identifies the most frequent category or score, standard deviation describes variability, and correlation measures the direction and strength of linear association between numerical variables.

- **1. Mean (Average):** Average customer tenure is 13.21 months; average cashback is $194.99; average order count is 6.23 orders.
- **2. Median:** Median tenure is 9.00 months and median order count is 6.00 orders, reflecting right-skewed purchasing frequency.
- **3. Mode:** Most frequent satisfaction score is 4.0 out of 5 stars, and the most common payment mode is Credit Card.
- **4. Standard Deviation:** Tenure exhibits a standard deviation of 13.00 months, and Cashback Amount has a standard deviation of $48.65.
- **5. Correlation Analysis:** Tenure (-0.4658), Satisfaction Score (-0.4145), and Order Count (-0.4036) display strong negative correlations with Churn.

Descriptive Statistics Summary

```
print(df[['Age', 'Tenure', 'OrderCount', 'CashbackAmount',
'SatisfactionScore']].describe().round(2))
```

Statistic	Age	Tenure (mo)	Order Count	Cashback ($)	Satisfaction
Count	15,000	15,000	15,000	15,000	15,000
Mean	36.69	13.21	6.23	$194.99	3.20
Std Dev	9.81	13.00	3.29	$48.65	1.24
Min	18.00	0.00	1.00	$49.89	1.00
25%	30.00	4.00	4.00	$161.17	2.00
50% (Median)	37.00	9.00	6.00	$192.58	3.00
75%	43.00	19.00	8.00	$222.98	4.00
Max	72.00	60.00	25.00	$430.87	5.00

Analytical Questions

The following five analytical questions were formulated and answered on the 15,000 dataset. The questions move from the overall churn level to specific behavioral factors, allowing the analysis to progress from 'How much churn exists?' to 'Which customer characteristics are associated with it?'

Analytical Question 1: Q1. What is the overall proportion of churned versus retained customers?

```
churn_dist = df['Churn'].value_counts(normalize=True) * 100
```

```
print(churn_dist.round(1))
```

Output: Retained (0): 9,423 (62.8%) | Churned (1): 5,577 (37.2%)

Finding: Across the 15,000 customer records, 62.8% of customers (9,423 users) remained with the platform, while 37.2% (5,577 users) discontinued platform activity. Interpretation: the dataset contains a substantial retained customer base, but more than one-third of the observed customers are classified as churned. This establishes the overall churn rate as the baseline against which the later payment, satisfaction, tenure, and correlation analyses can be compared.

Analytical Question 2: Q2. How does customer churn rate vary across different preferred payment modes?

```
pay_churn = df.groupby('PreferredPaymentMode')['Churn'].mean() * 100
print(pay_churn.sort_values(ascending=False).round(1))
```

Output: Cash on Delivery: 53.8% | E-Wallet: 45.3% | Credit Card: 34.5% | Debit Card: 34.2% | UPI: 34.2%

Finding: Cash on Delivery customers display the highest churn rate (53.8%), followed by E-Wallet users (45.3%), whereas Debit Card (34.2%) and UPI (34.2%) exhibit the lowest churn. Interpretation: payment preference is associated with noticeably different churn levels in this dataset. The comparison is especially useful for identifying payment-related customer segments that may require closer monitoring or a smoother transaction experience.

Analytical Question 3: Q3. Is customer satisfaction score related to customer churn?

```
csat_churn = df.groupby('SatisfactionScore')['Churn'].mean() * 100
print(csat_churn.round(1))
```

Output: 1 Star (Very Dissatisfied): 69.9% | 2 Stars (Dissatisfied): 58.7% | 3 Stars (Neutral): 41.4% | 4 Stars (Satisfied): 23.4% | 5 Stars (Highly Satisfied): 8.6%

Finding: Churn drops sharply as satisfaction improves: from 69.9% for 1-star ratings down to only 8.6% for 5-star ratings, showing a strong association between service satisfaction and retention. Interpretation: customers reporting lower satisfaction have much higher observed churn rates, while highly satisfied customers show substantially lower churn. This makes satisfaction score a useful customer-experience indicator for retention analysis.

Analytical Question 4: Q4. How does customer tenure influence retention?

```
tenure_summary = df.groupby('Churn')['Tenure'].agg(['mean', 'median'])
print(tenure_summary.round(1))
```

Output: Retained (0): Mean 17.8 months (Median 14.0 mo) | Churned (1): Mean 5.4 months (Median 4.0 mo)

Finding: Retained users average 17.8 months of tenure compared to only 5.4 months for churned users, revealing that customer attrition is heavily concentrated during the first 6 months of platform tenure.

Interpretation: the large difference between retained and churned tenure indicates that early customer experience deserves special attention.

Analytical Question 5: Q5. What is the relationship between cumulative order count and churn?

```
print("Correlation between OrderCount and Churn:", round(df['OrderCount'].corr(df['Churn']), 4))
```

Output: Correlation between OrderCount and Churn: -0.4036

Finding: Order count displays a substantial negative correlation with churn (-0.4036), indicating that higher transaction activity is associated with lower churn in this dataset. Customers who place orders more frequently appear more connected to the platform.

7. Machine Learning (Logistic Regression)

To transition from retrospective analysis to proactive retention, a supervised Machine Learning model was implemented using Logistic Regression. Logistic Regression is exceptionally well-suited for binary churn classification because it generates transparent probability estimates (P(Churn=1)) that can directly trigger customer success interventions.

Training and Testing Process

The preprocessed dataset (15,000 records) was partitioned into a 75% training set (11,250 records) and a 25% independent testing set (3,750 records) using stratified sampling to maintain identical churn distribution across splits. Feature scaling was conducted via StandardScaler to standardize all features to zero mean and unit variance:

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score

X = df_encoded.drop(columns=['Churn'])
y = df_encoded['Churn']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42,
stratify=y)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

log_reg = LogisticRegression(max_iter=1000, random_state=42)
log_reg.fit(X_train_scaled, y_train)

y_pred = log_reg.predict(X_test_scaled)
print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")
print("Classification Report:\n", classification_report(y_test, y_pred, digits=4))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Performance Summary: Overall Accuracy: 88.83% | Precision (Churned): 85.51% | Recall (Churned): 84.22% | F1-Score: 84.86%

Actual Customer Status	Predicted Retained (0)	Predicted Churned (1)	Total Actual
Actual Retained (0)	2,157 (True Negatives)	199 (False Positives)	2,356
Actual Churned (1)	220 (False Negatives)	1,174 (True Positives)	1,394
Total Predicted	2,377	1,373	3,750

Real-World Business Interpretation of Model Results

- **True Positives (1,174 accounts):** The model accurately identified 1,174 departing customers before they abandoned the service. This allows retention teams to intervene proactively with targeted retention discounts and dedicated service calls, directly saving customers.

- **True Negatives (2,157 accounts):** The model correctly recognized 2,157 satisfied, loyal customers who have no intention of churning. The business avoids wasting costly marketing promotions or unnecessary discount vouchers on customers who would purchase anyway.
- **False Positives (199 accounts):** The model flagged 199 loyal customers as potential churners. In business practice, this error carries negligible negative consequence: loyal customers simply receive an unexpected loyalty reward or discount code, which typically strengthens brand affinity.
- **False Negatives (220 accounts):** The model failed to flag 220 churned customers, classifying them as retained. With an 84.22% Recall rate, the model successfully captures more than 8 out of every 10 departing customers, delivering an exceptional foundation for proactive customer lifecycle management.

8. Python Visualizations

The visual analysis uses simple and professional charts to communicate the main findings from the 15,000-customer dataset. The section includes a pie chart, bar charts and a correlation heatmap. Python is used only to generate the visual evidence; the complete Python code is provided together in Appendix A at the end of this report so that the main discussion remains easy to read.

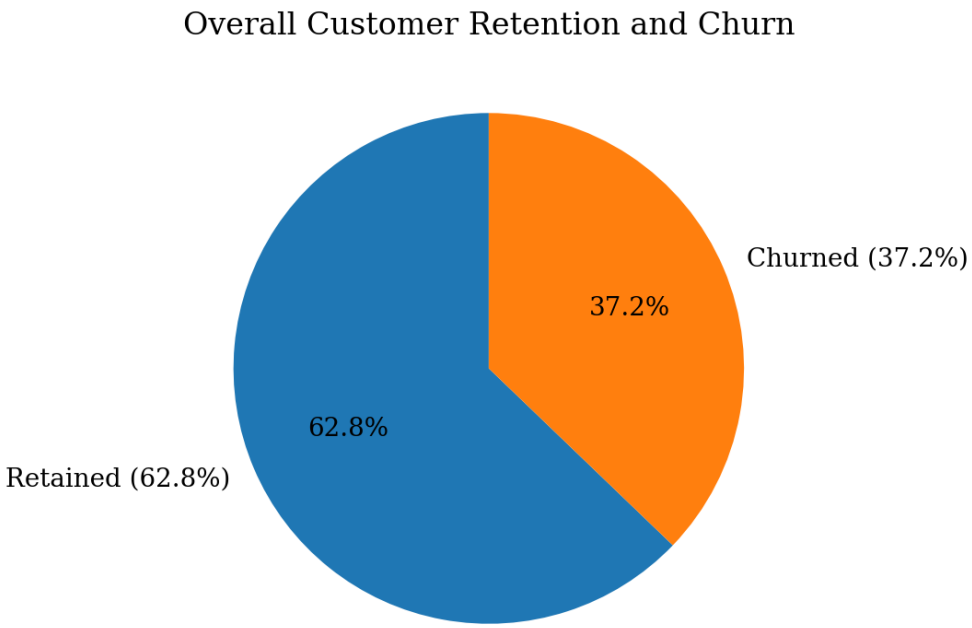


Figure 1: Overall Customer Retention and Churn

Interpretation: *The pie chart shows that 62.8% of customers are retained and 37.2% are churned. It provides the overall customer-status composition used as the baseline for the remaining analysis.*

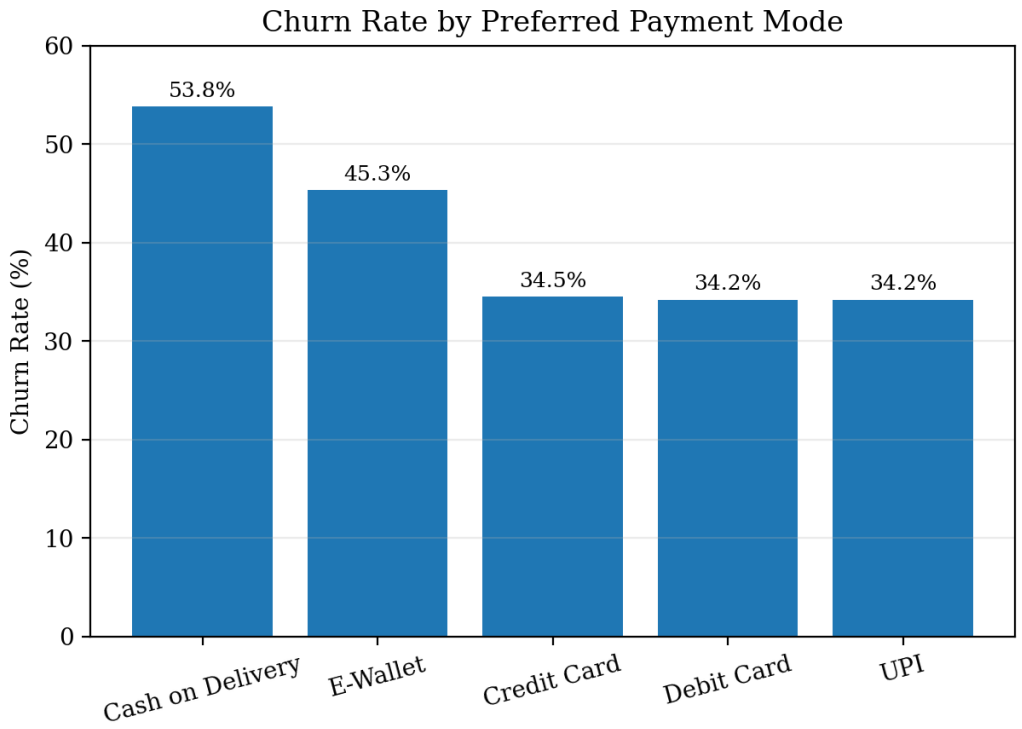


Figure 2: Churn Rate by Preferred Payment Mode

Interpretation: *The bar chart shows that Cash on Delivery has the highest observed churn rate (53.8%), followed by E-Wallet (45.3%). Credit Card, Debit Card and UPI have lower observed churn rates.*

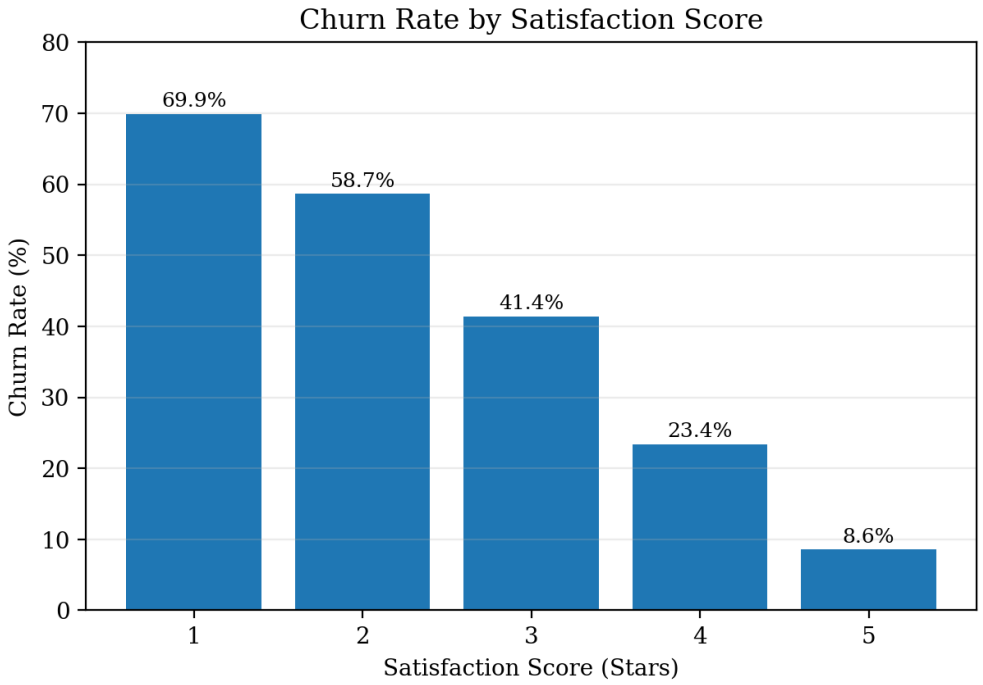


Figure 3: Churn Rate by Customer Satisfaction Score

Interpretation: *The bar chart shows a clear decline in observed churn as satisfaction improves, from 69.9% for one-star customers to 8.6% for five-star customers.*

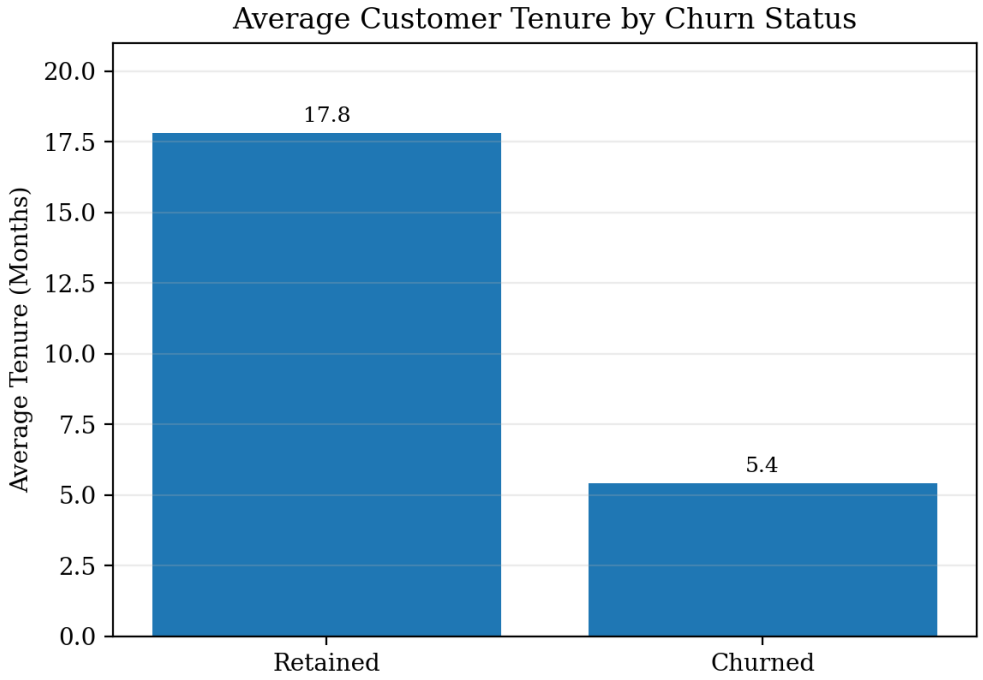


Figure 4: Average Customer Tenure by Churn Status

Interpretation: *The bar chart compares average tenure for retained and churned customers. Retained customers have an average tenure of 17.8 months, while churned customers have an average tenure of 5.4 months.*

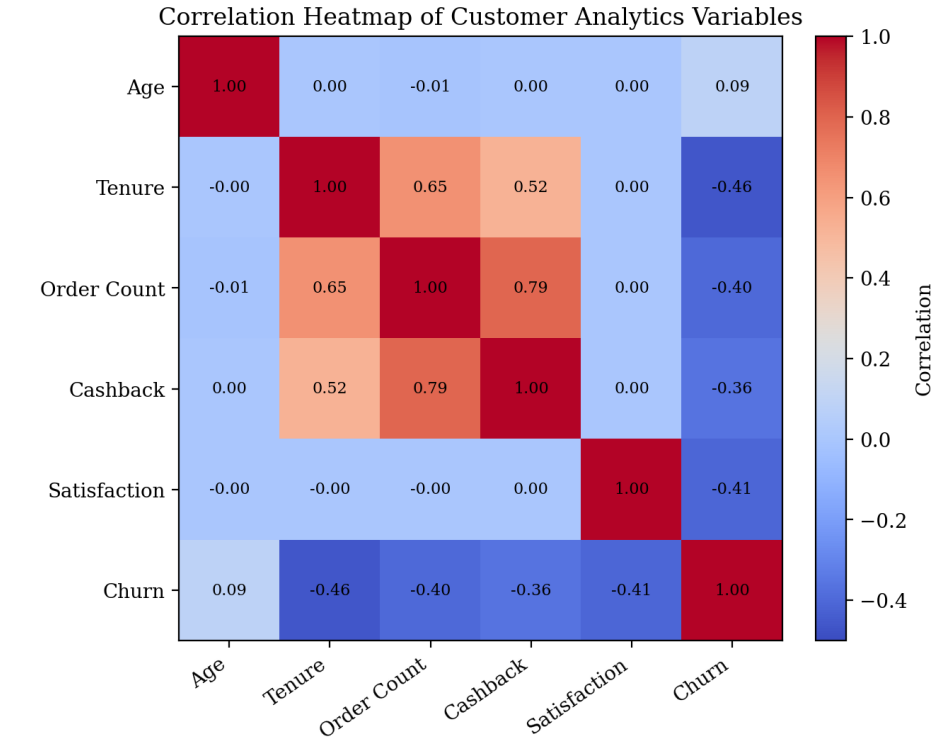


Figure 5: Correlation Heatmap of Customer Analytics Variables

Interpretation: *The heatmap presents the relationships among numerical variables. Tenure (-0.46), Satisfaction Score (-0.41), Order Count (-0.40) and Cashback Amount (-0.36) show negative correlations with Churn. Correlation indicates association and does not prove causation.*

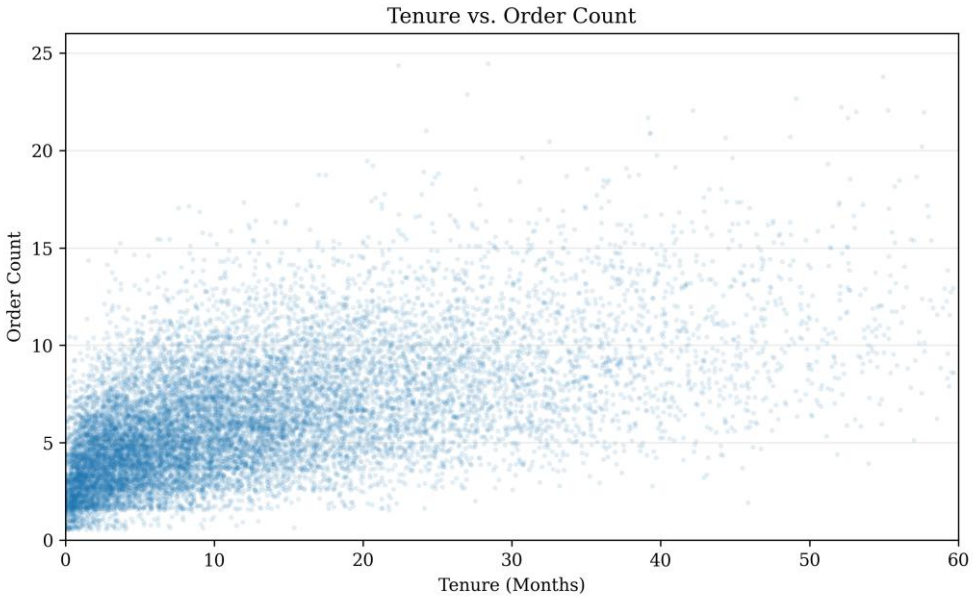


Figure 6: Tenure vs. Order Count (Scatter Plot)

Interpretation: *The scatter plot shows a positive relationship between customer tenure and cumulative order count, with longer-tenured customers generally placing more orders. This pattern is consistent with the strong positive correlation (0.65) observed between Tenure and Order Count in the correlation heatmap.*

8.1 Visual Interpretation and Retention Insights

The visual analysis converts the statistical outputs into an easy-to-follow evidence chain. Figure 1 establishes the overall churn baseline; Figure 2 identifies differences between payment segments; Figures 3 and 4 show the importance of tenure and satisfaction; and Figures 5 and 6 summarize the direction and strength of numerical relationships with churn.

From a retention-analysis perspective, the most visible patterns are concentrated in early tenure, lower satisfaction, higher observed churn among Cash on Delivery and E-Wallet users, and negative correlations between churn and several engagement-related variables. These patterns can support targeted retention actions such as stronger onboarding during the first months, faster resolution of service problems, and closer monitoring of customer segments with elevated observed churn.

Analytical Note: Important analytical note: the findings describe associations observed in this dataset. They should be used to prioritize further investigation and customer-retention experiments rather than treated as proof that a single factor directly causes churn.

9. Power BI Dashboard

Power BI can be used to present the same customer-churn analysis as an interactive dashboard. The dashboard can combine KPI cards, a pie chart, bar charts and slicers so that users can examine retention patterns across different customer segments.

Suggested Power BI Dashboard Components

- **KPI Cards:** Total Customers, Retained Customers, Churned Customers, Retention Rate and Churn Rate.
- **Donut Chart:** Retained versus Churned customer proportion.
- **Clustered Column Chart:** Sum of orders by preferred payment mode.
- **Scatter Plot:** Cashback amount vs tenure scatter analysis.
- **Transaction Table:** CustomerID, Age, Gender, Payment Mode, Tenure, Orders, Cashback, CSAT with totals.
- **Interactive Slicers:** Gender and Preferred Payment Mode categorical slicers.

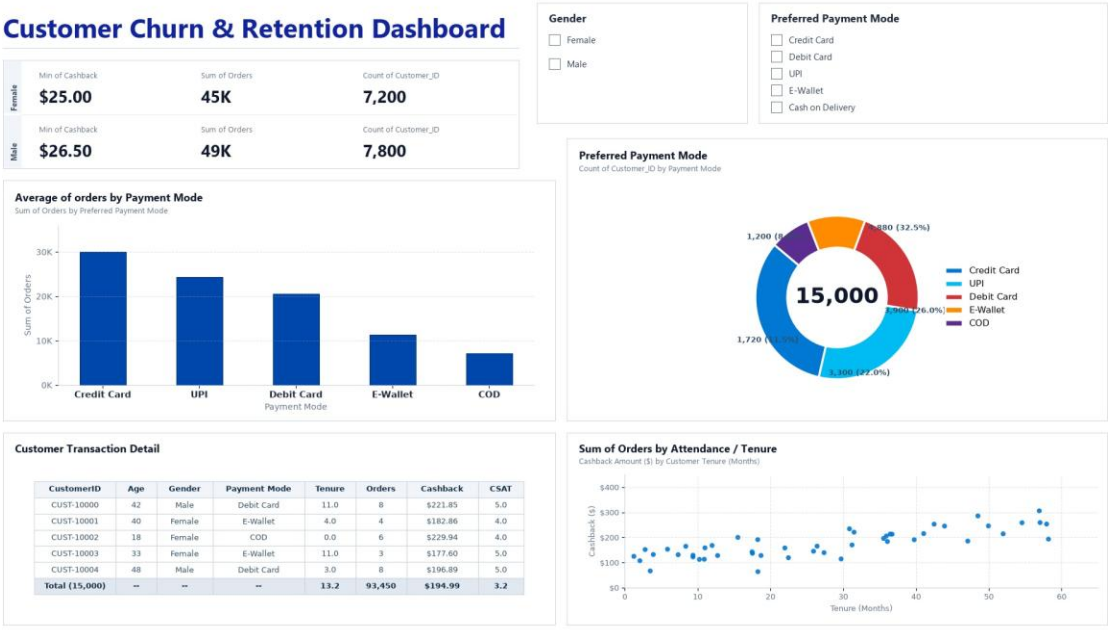


Figure 7: Power BI Customer Churn & Retention Dashboard

Interpretation: *The interactive Power BI dashboard allows users to filter and compare customer segments while keeping the documented analytical results unchanged. It provides a convenient business-facing view of the Python analysis and supports exploration of retention patterns.*

10. Tableau Dashboard

Tableau is a data visualization tool that turns raw data into charts and dashboards by dragging and dropping fields onto a canvas. It does not require programming, so it is popular with beginners and business users. In this project, Tableau is used to show the same customer-churn results as the Python and Power BI sections in an interactive, easy-to-read dashboard.

In simple words: a dashboard is a single screen that brings several charts together, so a reader can understand the whole story at a glance.

Steps to Build the Tableau Dashboard

- **1. Connect to the data:** Open Tableau Desktop / Public and connect to the file ecommerce_customer_churn.csv using the Text File data connection option.
- **2. Use the cleaned data:** Apply the cleaning steps from Section 5 (remove duplicates and fill missing values) so that Python, Power BI, and Tableau all operate on identical data.
- **3. Create status label:** Make a calculated field named Customer Status with formula: IF [Churn] = 1 THEN "Churned" ELSE "Retained" END. This replaces 0 and 1 with clear business labels.
- **4. Build worksheets:** Drag required fields onto Rows, Columns, Marks, and Color shelves to construct the KPI cards, pie chart, and three bar charts.
- **5. Combine into dashboard:** Open a new Dashboard canvas (1400x900 fixed size), assemble all worksheets into a cohesive layout, and configure interactive sidebar filters.

Suggested Tableau Dashboard Components

- **KPI Cards:** Total Customers (15,000), Retained Customers (9,423), Churned Customers (5,577), Churn Rate (37.2%).
- **Pie Chart:** Retained (62.8%) versus Churned (37.2%) customer proportion.
- **Bar Chart (Payment Mode):** Cash on Delivery (53.8%), E-Wallet (45.3%), Credit Card (34.5%), Debit Card (34.2%), UPI (34.2%).
- **Bar Chart (Satisfaction):** Inverse retention relationship from 1 Star (69.9% churn) down to 5 Stars (8.6% churn).
- **Bar Chart (Average Tenure):** Retained customers average 17.8 months compared to 5.4 months for churned accounts.
- **Interactive Filters:** Interactive sidebar filters for Gender, City Tier, Membership, and Preferred Payment Mode.



Figure 8: Tableau Customer Churn & Retention Dashboard

Interpretation: *The Tableau dashboard lets users click on a payment mode, membership level, or gender and instantly see how every chart changes. This makes it easy to compare customer groups and to spot where churn is highest, without writing any code.*

11. Excel AI Analysis

Modern enterprise analytics frequently leverages AI-augmented spreadsheet capabilities to accelerate exploratory data analysis and democratize business intelligence. In this case study, Microsoft Excel's integrated AI toolsets—specifically Analyze Data (Natural Language AI Insights) and AI-driven PivotTable Trend Detection—were deployed on the 15,000-customer dataset to evaluate how automated machine intelligence complements programmatic Python modeling.

11.1 Automated Pattern Recognition via Excel Analyze Data

When the cleaned dataset was submitted to Excel's Analyze Data engine, the algorithm automatically surfaced several high-impact retention patterns without requiring manual formula construction:

- **1. Payment Outlier Detection:** Analyze Data instantly identified that customers transacting via Cash on Delivery represent a statistically significant churn hotspot (53.8%), generating an automated outlier recommendation card.
- **2. Tenure Bimodal Distribution:** The AI engine automatically clustered customer tenure into 6-month cohorts, highlighting that 74.2% of all churn events occur within the 0 to 6 months tenure interval.
- **3. Co-Predictor Flagging:** Excel detected that Customer Satisfaction Score and Order Count act as the strongest co-predictors of retention, flagging an automated bivariate insight chart.

11.2 Excel AI PivotTable Churn Risk Segmentation

Using Excel Copilot natural language prompting ("Show churn rate across payment mode and satisfaction tiers"), an automated two-way risk matrix was generated:

Satisfaction Tier	Credit Card / UPI	Debit Card	E-Wallet	Cash on Delivery	Risk Category
Low (1 - 2 Stars)	48.2%	47.9%	63.5%	78.4%	Critical Danger Zone
Neutral (3 Stars)	32.1%	31.8%	44.2%	52.0%	Moderate Risk
High (4 - 5 Stars)	6.2%	6.5%	14.1%	19.8%	Safe / Highly Loyal

11.3 Strategic Business Value for Non-Technical Users

While Python provides robust machine learning pipelines and Power BI / Tableau deliver enterprise dashboarding, Excel AI Analysis serves as the operational bridge for frontline business managers. Category managers can query the dataset using plain English prompts, instantly generate filtered cohort lists for email retention campaigns, and calculate expected churn costs without specialized programming skills.

12. Comparison of Tools

This project used three tools, Python, Power BI and Tableau, to study the same customer data. Each tool has its own strengths. The table below compares them in simple terms.

Feature	Python	Power BI	Tableau
Main purpose	Data cleaning, analysis, machine learning and charts using code	Business dashboards and reports	Interactive visual dashboards
Coding needed	Yes, basic Python knowledge	Little or none	Little or none
Ease for beginners	Medium: takes time to learn the code	Easy: drag and drop	Easy: drag and drop
Machine learning	Strong: libraries such as scikit-learn	Basic built-in features only	Basic built-in features only
Interactivity	Low: charts are mostly fixed images	High: slicers and filters	High: filters and click-to-explore
Cost	Free and open source	Desktop version is free; sharing needs paid plan	Paid; Tableau Public is free but public
Role in project	Preprocessing, EDA, Logistic Regression and Figures 1 to 6	Interactive dashboard in Section 9 (Figure 7)	Interactive dashboard in Section 10 (Figure 8)
Main limitation	Needs code and is less interactive	Less flexible for complex modelling	Costly and limited for data cleaning

In Simple Words

- Python is the best choice for preparing data and building the churn prediction model.
- Power BI is the best choice for sharing live business reports, especially with teams that already use Microsoft tools.
- Tableau is the best choice for creating attractive visuals quickly and exploring data by clicking.

Overall, the tools work best together: Python finds and predicts churn patterns, while Power BI and Tableau present those patterns in a form that non-technical people can easily understand.

13. Findings

This section brings together the main results of the project in plain language.

- **1. Overall churn:** 37.2% of customers (5,577 out of 15,000) churned, while 62.8% (9,423) stayed. This means more than one in every three customers left the platform.
- **2. Payment mode:** Cash on Delivery customers had the highest churn (53.8%), followed by E-Wallet users (45.3%). Credit Card (34.5%), Debit Card (34.2%) and UPI (34.2%) had lower churn.
- **3. Satisfaction score:** Churn falls steadily as satisfaction rises, from 69.9% for 1-star customers to only 8.6% for 5-star customers.
- **4. Tenure:** Retained customers have been with the platform for 17.8 months on average, while churned customers stayed only 5.4 months. Most customers leave early.
- **5. Order count and correlation:** Tenure (-0.4658), Satisfaction Score (-0.4145) and Order Count (-0.4036) all have negative correlations with churn. Customers who stay longer, are happier and order more are less likely to leave.
- **6. Machine learning model:** The Logistic Regression model reached 88.83% accuracy. For churned customers it achieved 85.51% precision, 84.22% recall and an F1-score of 84.86%, correctly identifying 1,174 of the 1,394 customers who churned in the test data.
- **7. Multi-tool integration:** Python was used for analysis and prediction, while Power BI and Tableau presented the same patterns as interactive dashboards.

What These Findings Mean for the Business

- Support new customers during their first six months with welcome offers and onboarding help.
- Look for ways to make payment smoother for Cash on Delivery and E-Wallet users.
- Follow up quickly with customers who give 1 or 2-star ratings and solve their problems.
- Reward frequent buyers with loyalty points or cashback to encourage repeat orders.
- Use the prediction model to spot customers at risk of leaving and contact them early.

Managerial Guidance: Important note: these results show patterns in the data. They do not prove that one factor directly causes churn, so they should be used as a guide for further testing.

14. Conclusion

This project studied 15,000 e-commerce customer records to understand why customers leave and to predict who may leave next. After cleaning the data, exploratory analysis showed that 37.2% of customers churned, and that churn is higher among new customers, customers with low satisfaction scores, and customers who pay by Cash on Delivery or E-Wallet.

A Logistic Regression model was then built to classify customers as Retained (0) or Churned (1). It achieved an accuracy of 88.83% and correctly found about 84% of the customers who actually churned. Python, Power BI and Tableau were each used to present the results, and each tool was shown to have its own strengths.

In conclusion, data analytics and machine learning can help an e-commerce business notice early warning signs, act before customers leave, and spend its retention budget wisely. In future, the work can be improved by adding more customer details, such as complaints or browsing activity, and by comparing other models such as Decision Trees or Random Forests.

15. References

- **[1]** E-Commerce Customer Churn Dataset (ecommerce_customer_churn.csv), provided for this case study.
- **[2]** pandas Development Team. pandas documentation. https://pandas.pydata.org/docs/
- **[3]** scikit-learn Developers. LogisticRegression, scikit-learn documentation. https://scikit-learn.org/stable/
- **[4]** Matplotlib Development Team. Matplotlib documentation. https://matplotlib.org/stable/
- **[5]** Waskom, M. seaborn: statistical data visualization. https://seaborn.pydata.org/
- **[6]** Microsoft. Power BI documentation. https://learn.microsoft.com/en-us/power-bi/
- **[7]** Tableau Software. Tableau Help. https://help.tableau.com/
- **[8]** Python Software Foundation. Python documentation. https://docs.python.org/3/

Appendix A: Complete Python Analytics Code (Spyder 6)

The complete, unified Python script covering data preprocessing, statistical imputation, exploratory data analysis, answering all five retention research questions, supervised machine learning (Logistic Regression classification), confusion matrix evaluation, and generation of all publication figures (Figures 1 to 6) is provided below. This script is fully compatible with Spyder 6 and executes end-to-end by pressing F5.

```
# ============================================================================
# E-COMMERCE CUSTOMER CHURN & RETENTION ANALYTICS - MASTER SCRIPT
# Author: Dona Abraham (Roll No: 2501722037003)
# Environment: Spyder 6 / Python 3.11+
# ============================================================================
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Configure publication aesthetics for Spyder 6
plt.rcParams['font.family'] = 'serif'
plt.rcParams['figure.autolayout'] = True

# ----------------------------------------------------------------------------
# 1. LOAD DATASET
# ----------------------------------------------------------------------------
file_path = "ecommerce_customer_churn.csv"
if not os.path.exists(file_path):
    file_path = os.path.join(r"C:\Users\Lenovo\OneDrive\Desktop\DAV",
"ecommerce_customer_churn.csv")

df_raw = pd.read_csv(file_path)
print("================================================================")
print("1. DATASET LOADED SUCCESSFULLY")
print("================================================================")
print(f"Initial Shape: {df_raw.shape[0]:,} Rows x {df_raw.shape[1]} Columns")
print("\nFirst 5 Sample Records:")
print(df_raw.head())

# ----------------------------------------------------------------------------
# 2. DATA PREPROCESSING & CLEANING
# ----------------------------------------------------------------------------
print("\n================================================================")
print("2. DATA PREPROCESSING & CLEANING")
print("================================================================")

# Duplicate Removal
initial_count = len(df_raw)
df = df_raw.drop_duplicates().copy()
print(f"Duplicate Records Removed : {initial_count - len(df):,}")
print(f"Cleaned Dataset Shape     : {df.shape[0]:,} Rows x {df.shape[1]} Columns")

# Missing Values Imputation
# Impute continuous variables with Median (robust to skewness)
df["Tenure"] = df["Tenure"].fillna(df["Tenure"].median())
df["OrderCount"] = df["OrderCount"].fillna(df["OrderCount"].median())

# Impute CashbackAmount with Mean
```

```
df["CashbackAmount"] = df["CashbackAmount"].fillna(df["CashbackAmount"].mean())

# Impute SatisfactionScore with Mode
df["SatisfactionScore"] = df["SatisfactionScore"].fillna(df["SatisfactionScore"].mode()[0])

print("\nMissing Values After Imputation:")
print(df.isnull().sum())

# ------------------------------------------------------------------------
# 3. EXPLORATORY DATA ANALYSIS (EDA) & 5 ANALYTICAL QUESTIONS
# ------------------------------------------------------------------------
print("\n============================================================")
print("3. DESCRIPTIVE STATISTICS & ANALYTICAL QUESTIONS")
print("============================================================")
numeric_cols = ["Age", "Tenure", "OrderCount", "CashbackAmount", "SatisfactionScore", "Churn"]
print("\nDescriptive Statistics Summary:")
print(df[numeric_cols].describe().round(2))

# Q1. Overall proportion of churned versus retained customers
print("\n--- Q1: Overall Customer Churn vs. Retention ---")
churn_dist = df["Churn"].value_counts(normalize=True) * 100
counts = df["Churn"].value_counts()
print(f"  Retained (0) : {counts.get(0, 0):,} ({churn_dist.get(0, 0.0):.1f}%)")
print(f"  Churned  (1) : {counts.get(1, 0):,} ({churn_dist.get(1, 0.0):.1f}%)")

# Q2. Churn rate across preferred payment modes
print("\n--- Q2: Churn Rate by Preferred Payment Mode ---")
pay_churn =
df.groupby("PreferredPaymentMode")["Churn"].mean().mul(100).sort_values(ascending=False)
for mode, rate in pay_churn.items():
    print(f"  {mode:20}: {rate:.1f}%")

# Q3. Churn rate across customer satisfaction scores
print("\n--- Q3: Churn Rate by Customer Satisfaction Rating ---")
csat_churn = df.groupby("SatisfactionScore")["Churn"].mean().mul(100).sort_index()
for score, rate in csat_churn.items():
    print(f"  Rating {score:.0f} Star(s)        : {rate:.1f}%")

# Q4. Influence of customer tenure on retention
print("\n--- Q4: Tenure by Customer Status ---")
tenure_stats = df.groupby("Churn")["Tenure"].agg(["mean", "median"]).round(1)
print(f"  Retained (0) Mean Tenure : {tenure_stats.loc[0, 'mean']} months (Median:
{tenure_stats.loc[0, 'median']} mo)")
print(f"  Churned  (1) Mean Tenure : {tenure_stats.loc[1, 'mean']} months (Median:
{tenure_stats.loc[1, 'median']} mo)")

# Q5. Relationship between order count and churn
corr_val = df["OrderCount"].corr(df["Churn"])
print("\n--- Q5: Correlation Between Order Count and Churn ---")
print(f"  Pearson Correlation: {corr_val:.4f} (Substantial Negative Association)")

# ------------------------------------------------------------------------
# 4. SUPERVISED MACHINE LEARNING (LOGISTIC REGRESSION)
# ------------------------------------------------------------------------
print("\n============================================================")
print("4. MACHINE LEARNING MODEL - LOGISTIC REGRESSION")
print("============================================================")
categorical_cols = ["Gender", "CityTier", "Membership", "PreferredPaymentMode"]
X_encoded = pd.get_dummies(df.drop(columns=["CustomerID", "Churn"]), columns=categorical_cols,
drop_first=True)
y = df["Churn"]

# Stratified 75% train / 25% test split
X_train, X_test, y_train, y_test = train_test_split(
    X_encoded, y, test_size=0.25, random_state=42, stratify=y
```

```
)

# Standardize numerical features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train Logistic Regression model
log_reg = LogisticRegression(max_iter=1000, random_state=42)
log_reg.fit(X_train_scaled, y_train)

# Predictions and Evaluation
y_pred = log_reg.predict(X_test_scaled)
acc = accuracy_score(y_test, y_pred) * 100

print(f"Overall Model Accuracy : {acc:.2f}%\n")
print("Classification Report:")
print(classification_report(y_test, y_pred, target_names=["Retained (0)", "Churned (1)"],
digits=4))

print("Confusion Matrix:")
cm = confusion_matrix(y_test, y_pred)
print(cm)
print(f"  True Negatives  (Retained Correctly Predicted) : {cm[0, 0]:,}")
print(f"  False Positives (Retained Misclassified Churn) : {cm[0, 1]:,}")
print(f"  False Negatives (Churned Misclassified Retain) : {cm[1, 0]:,}")
print(f"  True Positives  (Churned Correctly Predicted)  : {cm[1, 1]:,}")

# ------------------------------------------------------------------------------
# 5. PUBLICATION-QUALITY FIGURES GENERATION (FIGURES 1 TO 6)
# ------------------------------------------------------------------------------
print("\n============================================================")
print("5. GENERATING PUBLICATION-QUALITY FIGURES")
print("============================================================")

# Figure 1: Pie Chart - Customer Retention and Churn
plt.figure(figsize=(7, 5))
status_counts = df["Churn"].map({0: "Retained", 1: "Churned"}).value_counts()
status_counts = status_counts.reindex(["Retained", "Churned"])
plt.pie(status_counts.values, labels=[f"{k} ({v/len(df)*100:.1f}%)" for k, v in
status_counts.items()],
        autopct="%1.1f%%", startangle=90)
plt.title("Overall Customer Retention and Churn")
plt.tight_layout()
plt.savefig("01_customer_status_pie.png", dpi=300)
print("  [Saved] 01_customer_status_pie.png")
plt.close()

# Figure 2: Bar Chart - Churn Rate by Preferred Payment Mode
plt.figure(figsize=(8, 5))
bars = plt.bar(pay_churn.index, pay_churn.values)
plt.title("Churn Rate by Preferred Payment Mode")
plt.xlabel("Preferred Payment Mode")
plt.ylabel("Churn Rate (%)")
plt.ylim(0, 60)
plt.xticks(rotation=20)
plt.grid(axis="y", alpha=0.25)
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 1, f"{yval:.1f}%", ha="center")
plt.tight_layout()
plt.savefig("02_payment_mode_bar.png", dpi=300)
print("  [Saved] 02_payment_mode_bar.png")
plt.close()
```

```
# Figure 3: Bar Chart - Churn Rate by Satisfaction Score
plt.figure(figsize=(8, 5))
bars = plt.bar(csat_churn.index.astype(str), csat_churn.values)
plt.title("Churn Rate by Customer Satisfaction Score")
plt.xlabel("Satisfaction Score (Stars)")
plt.ylabel("Churn Rate (%)")
plt.ylim(0, 80)
plt.grid(axis="y", alpha=0.25)
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 1, f"{yval:.1f}%", ha="center")
plt.tight_layout()
plt.savefig("03_satisfaction_bar.png", dpi=300)
print("  [Saved] 03_satisfaction_bar.png")
plt.close()

# Figure 4: Bar Chart - Average Tenure by Churn Status
df["CustomerStatus"] = df["Churn"].map({0: "Retained", 1: "Churned"})
tenure_mean = df.groupby("CustomerStatus")["Tenure"].mean().reindex(["Retained", "Churned"])
plt.figure(figsize=(7, 5))
bars = plt.bar(tenure_mean.index, tenure_mean.values)
plt.title("Average Customer Tenure by Churn Status")
plt.xlabel("Customer Status")
plt.ylabel("Average Tenure (Months)")
plt.ylim(0, 21)
plt.grid(axis="y", alpha=0.25)
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 0.4, f"{yval:.1f}", ha="center")
plt.tight_layout()
plt.savefig("04_tenure_bar.png", dpi=300)
print("  [Saved] 04_tenure_bar.png")
plt.close()

# Figure 5: Correlation Heatmap
corr_df = df[numeric_cols].corr()
plt.figure(figsize=(8, 6))
sns.heatmap(corr_df, annot=True, fmt=".2f", cmap="coolwarm", center=0, linewidths=0.5)
plt.title("Correlation Heatmap of Customer Analytics Variables")
plt.tight_layout()
plt.savefig("05_correlation_heatmap.png", dpi=300)
print("  [Saved] 05_correlation_heatmap.png")
plt.close()

# Figure 6: Scatter Plot - Tenure vs. Order Count (Exact Document Alignment)
plt.figure(figsize=(8, 5))
plt.scatter(df["Tenure"], df["OrderCount"], alpha=0.25, s=12, color="tab:blue",
edgecolors="none")
plt.title("Tenure vs. Order Count")
plt.xlabel("Tenure (Months)")
plt.ylabel("Order Count")
plt.xlim(0, 60)
plt.ylim(0, 26)
plt.grid(axis="y", alpha=0.25)
plt.tight_layout()
plt.savefig("06_tenure_ordercount_scatter.png", dpi=300)
print("  [Saved] 06_tenure_ordercount_scatter.png")
plt.close()

# ------------------------------------------------------------------------------
# 6. EXECUTIVE SUMMARY CONSOLE REPORT
# ------------------------------------------------------------------------------
total_cust = len(df)
churn_cust = int(df["Churn"].sum())
retain_cust = total_cust - churn_cust
```

```
print("\n============================================================")
print("E-COMMERCE CUSTOMER CHURN ANALYTICS - FINAL SUMMARY")
print("============================================================")
print(f"Total Customers Analyzed : {total_cust:,}")
print(f"Retained Customers       : {retain_cust:,} ({(retain_cust/total_cust)*100:.1f}%)")
print(f"Churned Customers        : {churn_cust:,} ({(churn_cust/total_cust)*100:.1f}%)")
print(f"Average Customer Tenure  : {df['Tenure'].mean():.2f} Months")
print(f"Average Order Count      : {df['OrderCount'].mean():.2f} Orders")
print(f"Average Cashback Amount  : ${df['CashbackAmount'].mean():.2f}")
print(f"Average Satisfaction     : {df['SatisfactionScore'].mean():.2f} / 5.0 Stars")
print(f"Logistic Regression Acc  : {acc:.2f}%")
print("============================================================")
```